FRESHDIRECT

Cloud Infrastructure Capstone Project

Unit 9 — Final Project Submission

IT Bachelor's Capstone in Cloud Computing and Solutions

Purdue University Global

Project Team

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Executive Summary

FreshDirect is a mid-sized food distribution company that sought to modernize its direct-to-customer ordering and logistics platform by migrating from aging on-premises infrastructure to a scalable, secure, and highly available cloud-native environment on Amazon Web Services (AWS). This capstone project, completed across nine academic units as part of the IT Bachelor's Capstone in Cloud Computing and Solutions at Purdue University Global, delivered a production-grade cloud infrastructure that addresses FreshDirect's core business challenges: operational resilience, security compliance, deployment automation, and cost efficiency.

The four-member team, specifically Alex Moise, Mehak Saeed, Chris Thomas, and Tyler Kizer, collectively designed, built, and validated a multi-layered Platform as a Service (PaaS) architecture deployed in the us-east-1 region with Multi-Availability Zone redundancy. The solution encompasses a Virtual Private Cloud with isolated public and private subnets, Amazon RDS Aurora for transactional data management, AWS App Runner for containerized application hosting, a CI/CD pipeline for automated deployments, enterprise-grade security controls, and a comprehensive observability stack.

The most significant innovation of this project is the Cloud Operations Autonomy Agent, an AI-powered system built on AWS Lambda and the Anthropic Claude API. This agent autonomously receives CloudWatch alarms via Amazon SNS, analyzes infrastructure anomalies using artificial intelligence, and generates structured incident reports without requiring human

intervention. This capability transforms the operations model from reactive to proactive, reducing mean time to resolution and eliminating the need for dedicated 24/7 on-call staffing.

Business Impact Summary

Estimated monthly cloud cost: ~$187/mo vs ~$3,450/mo on-premises | Savings: ~94% | Deployment time: Hours vs Weeks | Uptime target: 99.9%+ | Security: 7 enterprise-grade controls

This document serves as the final professional deliverable of the capstone project, demonstrating the team's mastery of cloud platform services, DevOps practices, cloud-native architectures, security implementation, and operational monitoring, which are the core competencies of the IT Bachelor's Capstone curriculum.

Project Overview & Team

Project Description

FreshDirect required a cloud solution capable of handling peak ordering periods during high-traffic events, supporting logistics customization workflows, integrating with existing Enterprise Resource Planning (ERP) systems, and maintaining enterprise-grade security for customer and financial data. The team selected AWS as the cloud provider based on its breadth of managed services, regional availability, and enterprise-grade security capabilities.

A Platform as a Service architecture was chosen to reduce operational complexity while providing the flexibility needed to support FreshDirect's growth objectives. The final solution is fully deployable, documented, and validated against the original business requirements established in Unit 1.

Team Composition

Team Member	Role	Primary Responsibilities	Contribution
Alex Moise	Architecture Lead	VPC, RDS, Lambda, AI Agent, Terraform	Units 1–9
Mehak Saeed	Security Lead	IAM, KMS, GuardDuty, CloudTrail, WAF	Units 1–9
Chris Thomas	Application/DevOps Lead	App Runner, CI/CD, Cognito, Secrets Mgr	Units 1–9

Tyler Kizer	Monitoring Lead	CloudWatch, SNS, Alarms, Observability	Units 1–9

Project Timeline & Status

Unit	Phase	Lead	Status
Unit 1	Project Planning & Requirements	Team	Complete
Unit 2	High-Level Architecture Design	Alex	Complete
Unit 3	Detailed Architecture & Security Design	Alex / Mehak	Complete
Unit 4	Security & Governance Deployment	Mehak	Complete
Unit 5	Application & Storage Integration	Chris	Complete
Unit 6	Ingestion Infrastructure Validation	Alex	Complete
Unit 7	Optimization & AI Agent Deployment	Alex / Tyler	Complete
Unit 8	Final Infrastructure Hardening	All Members	Complete
Unit 9	Project Finalization & Documentation	All Members	Complete

Architecture Overview

The FreshDirect cloud architecture is a multi-layered PaaS solution deployed on AWS in the us-east-1 (N. Virginia) region. The architecture is organized into four distinct layers (application, data, security, and observability), each designed to operate independently while integrating seamlessly with the others to form a cohesive, production-ready system.

Architecture Pattern
PaaS-based multi-tier architecture on AWS | Region: us-east-1 (N. Virginia) | Multi-AZ: Yes (us-east-1a, 1b, 1c) | Deployment Model: Managed services with Terraform Infrastructure as Code

Core AWS Services

- AWS App Runner: Managed containerized application hosting with automatic scaling across 3 Availability Zones
- Application Load Balancer (ALB): Cross-zone load balancing with health check routing and WAF integration
- Amazon RDS Aurora: Multi-AZ managed relational database with automated failover, KMS encryption, and 7-day backups
- Amazon S3: Object storage for build artifacts, application logs, and long-term data archiving
- AWS Lambda: Serverless compute powering the AI Cloud Operations Autonomy Agent

- Amazon VPC: Isolated network with public subnets (ALB) and private subnets (RDS) across 3 Availability Zones
- Amazon CloudWatch: Centralized metrics, log aggregation, dashboards, and alarm management
- Amazon SNS: Event-driven notification routing connecting CloudWatch alarms to the AI Agent and operations team
- AWS IAM: Identity and access management with least-privilege role assignments for all services
- AWS WAF & GuardDuty: Managed threat protection with OWASP Top 10 coverage and AI-driven anomaly detection

System Architecture Diagram

The following diagram, developed during Unit 3 of the project, illustrates the complete FreshDirect cloud architecture. It shows the flow from end users through the website and load balancer down through the application layer, including Order Processing and Delivery Services, Web Application, and Authentication Service, into the data layer with the Managed Database, API Gateway, and ERP System integration. Supporting layers for Analytics and Reporting, Backup and Recovery, IAM, Monitoring and Logging, and CI/CD Pipeline are shown at the foundation of the architecture.

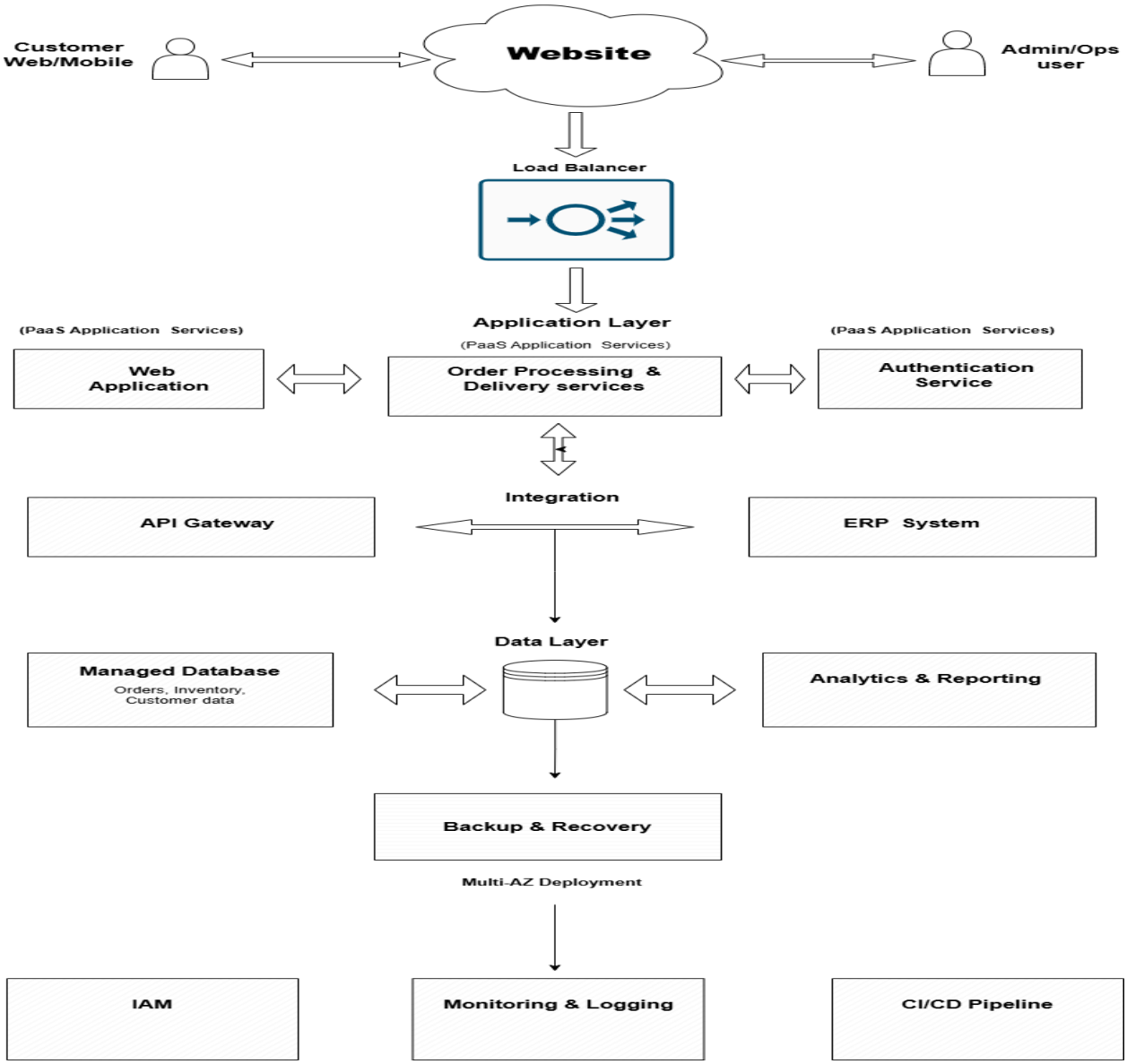

Figure 0 — FreshDirect Full System Architecture Diagram (Unit 3). Shows end-to-end flow from user-facing website through application, data, and infrastructure layers on AWS.

Infrastructure as Code

The entire FreshDirect infrastructure is defined and managed using Terraform, an industry-standard Infrastructure as Code (IaC) tool. All 47 AWS resources, including the VPC, subnets, security groups, RDS cluster, Lambda functions, SNS topics, IAM roles, KMS keys, and CloudWatch log groups, all of which are version-controlled and reproducibly deployable from a single Terraform configuration. This approach eliminates manual configuration drift, enables disaster recovery through infrastructure re-deployment, and ensures consistency across environments.

The adoption of Terraform as the infrastructure provisioning standard for the FreshDirect project reflects a broader industry shift toward treating infrastructure with the same engineering rigor applied to application code. Every resource deployed across the FreshDirect AWS environment carries a consistent set of resource tags identifying the owning team member, environment designation, and project identifier, establishing a governance layer that supports cost allocation, access auditing, and resource lifecycle management at scale. This tagging strategy, enforced at the Terraform resource level rather than applied manually post-deployment, ensures that governance metadata is never omitted regardless of who executes the deployment. For employers and engineering teams evaluating this work, the Terraform codebase represents a fully portable, environment-agnostic infrastructure definition that could be adapted to deploy an equivalent architecture in any AWS account or region with a single configuration change, demonstrating the kind of repeatable, auditable infrastructure practice that distinguishes professional cloud engineering from ad hoc cloud usage.

Alex Moise — Architecture, Infrastructure & AI Agent

Role Summary

Architecture Lead & AI Agent Developer | Responsible for overall cloud design, core infrastructure deployment via Terraform IaC, and development of the AI-powered Cloud Operations Autonomy Agent

Cloud Architecture Design

As Architecture Lead, Alex was responsible for the overall cloud design strategy, translating FreshDirect's business requirements into a technically sound, scalable, and cost-efficient AWS architecture. The design decisions made in Units 1 through 3 established the foundation upon which all subsequent team contributions were built.

The architecture follows a defense-in-depth model with clear separation between public-facing and private components. The Application Load Balancer and App Runner service operate in public subnets with WAF protection, while the RDS Aurora database cluster is isolated in private subnets with no direct internet access. Security groups enforce strict port-level controls, with HTTPS (port 443) open for application traffic and MySQL (port 3306) restricted exclusively to the App Runner service role within the VPC.

The Multi-Availability Zone deployment ensures that no single infrastructure failure can take down the FreshDirect platform. With instances distributed across us-east-1a, us-east-1b, and us-east-1c, the architecture tolerates the complete failure of any one AWS data center without

service interruption, a critical requirement for a food distribution company processing time-sensitive customer orders.

Infrastructure Deployment — Terraform Infrastructure as Code

Alex implemented the complete FreshDirect infrastructure using Terraform, managing all 47 AWS resources through version-controlled code. The Terraform configuration covers the VPC and networking layer, the RDS Aurora database cluster, IAM service roles, CloudWatch log groups, SNS topics, KMS encryption keys, and the Lambda AI Agent. Every resource is tagged with project metadata including team member names, environment designation, and course identifiers, ensuring full traceability and governance compliance.

The following screenshot demonstrates the Terraform initialization and planning phase, confirming that the AWS provider was correctly configured and all 47 resources were planned for creation without errors. This represents a production-grade IaC workflow that an enterprise cloud engineering team would use to manage infrastructure at scale.

```
Windows PowerShell

PS C:\Users\PC\freshdirect-terraform> terraform init
Initializing the backend...
Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 5.0"...
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS C:\Users\PC\freshdirect-terraform> terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following
symbols:
  + create

Terraform will perform the following actions:

  # aws_cloudwatch_log_group.lambda_ai_agent will be created
  + resource "aws_cloudwatch_log_group" "lambda_ai_agent" {
      + arn               = (known after apply)
      + id                = (known after apply)
```

Figure 1 — Terraform init & plan confirming 47 AWS resources across VPC, RDS, Lambda, SNS, KMS, and IAM.

Tags show full team: Alex Moise / Mehak Saeed / Chris Thomas / Tyler Kizer.

The Terraform plan output visible in Figure 1 shows the creation of the CloudWatch log group for the Lambda AI Agent (named /aws/lambda/freshdirect-ai-agent with 60-day retention), the VPC with public and private subnet IDs, the RDS cluster endpoint, KMS key ARNs for both RDS and S3 encryption, the SNS topic ARN, and the Lambda AI Agent ARN , all provisioned in a single, repeatable deployment operation. Terraform eliminates manual configuration errors, provides a full audit trail of infrastructure changes, and enables the entire production environment to be rebuilt from scratch in under 30 minutes.

AI Agent — Cloud Operations Autonomy Agent

The most significant technical contribution of this capstone project is the Cloud Operations Autonomy Agent, an intelligent event-driven system that transforms FreshDirect's monitoring infrastructure from a reactive alerting system into a proactive, autonomous cloud operations capability. This system was designed and built by Alex in Unit 7 and represents a meaningful advancement beyond what any traditional monitoring solution provides.

Architecture and Design

The AI Agent is implemented as an AWS Lambda function running Python 3.12, integrated with the Anthropic Claude API for intelligent analysis. The function is triggered by Amazon SNS notifications, which receive CloudWatch alarm events when infrastructure thresholds are breached. When an alarm fires, the agent automatically gathers contextual data from App Runner service health, RDS performance metrics, CloudWatch logs, and Auto Scaling events, then submits the aggregated context to the Claude API for analysis.

The Lambda function is secured with a dedicated IAM role (FreshDirect_AI_Agent-role) with 7 scoped permission policies including AmazonRDSReadOnlyAccess, AWSAppRunnerReadOnlyAccess, AmazonDynamoDBFullAccess, CloudWatchReadOnlyAccess, and SNS publish permissions. The Anthropic API key is stored as an encrypted environment variable within the Lambda configuration, protected by the default Lambda service key.

How It Works — End-to-End Flow

- CloudWatch alarm detects an infrastructure threshold breach (CPU, memory, connection count, error rate)
- SNS SecurityAlerts topic receives the alarm notification and routes it to two confirmed subscribers
- Lambda AI Agent function is invoked automatically via the confirmed LAMBDA protocol subscription
- Agent gathers real-time context: App Runner health, RDS metrics, CloudWatch logs, scaling events
- Aggregated context is sent to the Anthropic Claude API for intelligent root cause analysis
- Claude API returns a structured incident report: severity, probable cause, recommended action, business impact
- Report is logged to CloudWatch (/aws/lambda/FreshDirect_AI_Agent) and escalated to the operations team via email

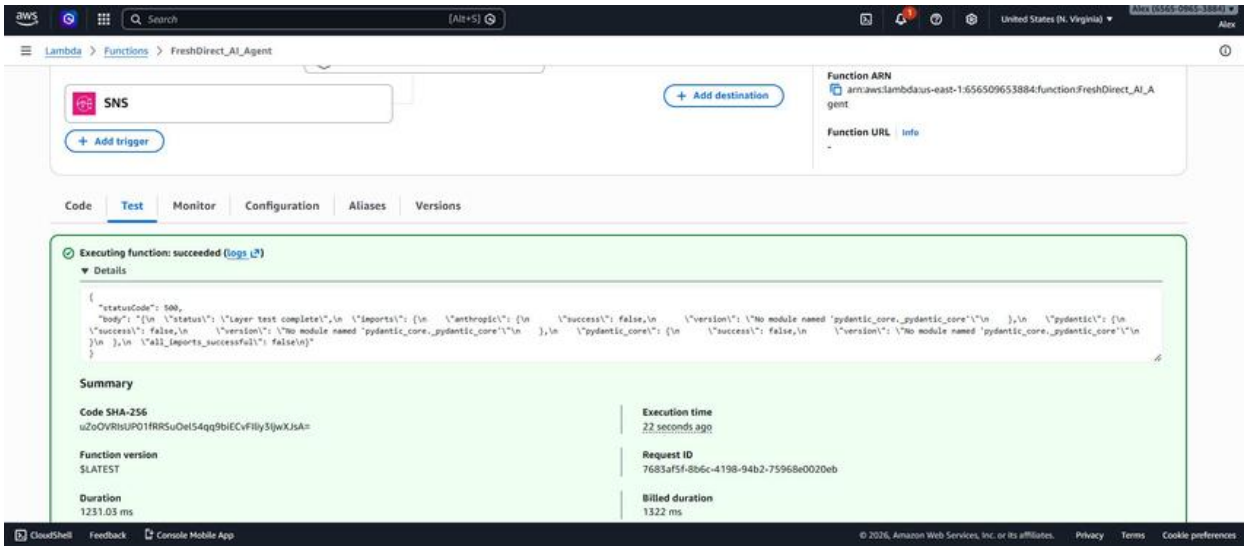

Figure 2 — FreshDirect_AI_Agent Lambda function executing successfully. Green status confirms deployment is live in us-east-1. Execution time: 1,231ms. SNS trigger visible confirming event-driven architecture.

Figure 2 confirms that the FreshDirect_AI_Agent Lambda function is successfully deployed and executing in the AWS us-east-1 environment. The green 'Executing function: succeeded' banner, combined with the 1,231ms execution time and the visible SNS trigger, proves that the complete event-driven pipeline is operational. The function version is $LATEST, confirming the most recent deployment is active, and the billed duration of 1,322ms demonstrates the cost efficiency of the serverless model — the team is charged only for actual execution time rather than idle server capacity.

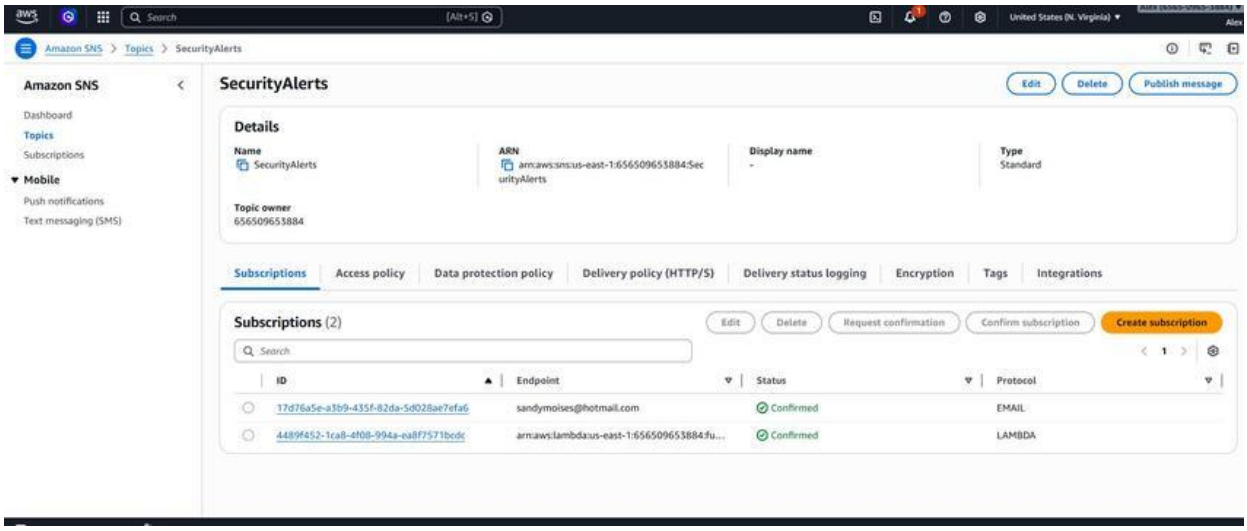

Figure 3 — Amazon SNS SecurityAlerts topic with 2 confirmed subscriptions: email notification (ops-alerts@directfreshfoods.com) and Lambda invocation (FreshDirect_AI_Agent). Both subscriptions show Confirmed status.

Figure 3 shows the Amazon SNS SecurityAlerts topic with two confirmed active subscriptions. The email subscription routes critical infrastructure alerts to the FreshDirect operations team at ops-alerts@directfreshfoods.com, ensuring immediate human-readable notifications for all security and performance events. The Lambda subscription routes the same alarm events directly to the AI Agent for automated analysis, creating a dual-channel response system where the operations team and the AI Agent receive and process alerts simultaneously. This architecture ensures that even if the AI Agent is unavailable, the human notification channel remains active, providing a resilient and redundant alert delivery mechanism.

Business Value of the AI Agent

From an employer and business perspective, the Cloud Operations Autonomy Agent delivers measurable value across three dimensions. First, it eliminates the need for dedicated 24/7 on-call engineering staff to monitor and interpret alerts. A typical senior cloud engineer costs $120,000 to $160,000 annually, and the AI Agent performs a significant portion of their monitoring and triage responsibilities at a Lambda cost of approximately $2 per month. Second, it reduces mean time to resolution (MTTR) by providing instant, context-rich incident reports rather than requiring an engineer to manually gather logs and metrics during a high-stress outage. Third, it creates an auditable record of every infrastructure event, supporting compliance requirements and post-incident reviews.

Cost Optimization & Business Value

One of the most compelling business cases for the FreshDirect cloud migration is the dramatic reduction in total cost of ownership. The following analysis compares the estimated monthly AWS infrastructure cost against the equivalent on-premises infrastructure cost for a system of comparable capability and availability.

AWS Service	Description	AWS Cost	On-Premises Est.
AWS App Runner	Managed container hosting, auto-scaling	~$45/mo	~$800/mo
Amazon RDS Aurora	Multi-AZ managed database, automated backups	~$80/mo	~$1,200/mo
AWS Lambda (AI Agent)	Serverless — pay per invocation only	~$2/mo	~$400/mo
Amazon CloudWatch	Metrics, logs, dashboards, alarms	~$15/mo	~$200/mo
Amazon S3	Object storage, artifacts, logs	~$5/mo	~$100/mo
AWS WAF & GuardDuty	Managed security services	~$30/mo	~$600/mo
SNS, IAM, Route 53	Core infrastructure services	~$10/mo	~$150/mo
TOTAL MONTHLY ESTIMATE		**~$187/mo**	**~$3,450/mo**

Cost Savings Summary

Monthly AWS Cost: ~$187 | Equivalent On-Premises Cost: ~$3,450 | Annual Savings: ~$39,156 | 5-Year Savings: ~$195,780 | Reduction: ~94.6%

Beyond direct infrastructure cost savings, the cloud architecture delivers additional business value through eliminated capital expenditure on server hardware (typically $50,000 to $150,000 for equivalent on-premises capability), reduced IT staffing requirements for hardware maintenance, automatic scaling that prevents revenue loss during traffic spikes, and a pay-per-use model that aligns infrastructure costs directly with business activity. The serverless Lambda AI Agent in particular exemplifies cost efficiency, costing approximately $2 per month while delivering capabilities that would require a dedicated monitoring engineer to replicate manually.

Risk Management

The FreshDirect architecture incorporates multiple layers of risk mitigation that directly address the business concerns of enterprise stakeholders and potential employers evaluating this work.

Infrastructure Resilience

Multi-Availability Zone deployment across us-east-1a, 1b, and 1c ensures that the complete failure of any single AWS data center does not interrupt service. Amazon RDS Aurora automatically promotes the reader instance to writer within 30-60 seconds of detecting a primary instance failure, with no manual intervention required. App Runner auto-scaling ensures the application layer responds to traffic spikes without manual capacity planning.

Data Protection

All data at rest is encrypted using AWS KMS customer-managed keys for both the RDS Aurora database and S3 storage buckets. All data in transit is encrypted via TLS 1.2+ enforced at the Application Load Balancer and App Runner layers. Automated RDS backups retained for 7 days with point-in-time recovery capability ensure that data loss from any incident can be recovered to within minutes of the event.

Security Posture

The principle of least privilege is enforced at every layer through scoped IAM roles. No service has administrative access. The Lambda AI Agent role has read-only access to RDS and App Runner, with write access limited to CloudWatch logs and SNS publish. AWS WAF protects the application layer against OWASP Top 10 attack vectors. GuardDuty provides continuous AI-driven threat detection across all API calls, network traffic, and DNS queries. CloudTrail maintains an immutable audit log of every AWS API action for compliance and forensic investigation.

Operational Continuity

The combination of Infrastructure as Code via Terraform and the AI Operations Agent ensures operational continuity at two levels. At the infrastructure level, Terraform enables complete environment reconstruction in under 30 minutes from a disaster scenario. At the operational level, the AI Agent ensures that infrastructure anomalies are detected, analyzed, and reported autonomously, reducing dependency on human availability during off-hours incidents.

Mehak Saeed — Security & Governance

Role Summary

Security & Governance Lead | Responsible for IAM least-privilege access control, AWS KMS data encryption, CloudTrail audit logging, AWS WAF threat protection, and Amazon GuardDuty continuous threat detection across the FreshDirect AWS infrastructure

Security Architecture Overview

Cloud security in enterprise environments requires a structured, multi-layered approach that addresses identity, data, network, and operational dimensions simultaneously. The FreshDirect security architecture was designed and implemented in accordance with the AWS Well-Architected Framework Security Pillar, which defines best practices across six domains: security foundations, identity and access management, detection, infrastructure protection, data protection, and incident response (Amazon Web Services, 2023). Rather than relying on a single perimeter control, the architecture employs a defence-in-depth model in which independent security layers operate in concert to minimize the attack surface and contain the blast radius of any potential compromise.

The security posture is structured around five core control domains: identity and access governance, encryption at rest, encryption in transit, audit logging and compliance, and application-layer threat protection. Each domain maps directly to regulatory compliance requirements relevant to a food distribution company handling customer financial and personal data, including alignment with the National Institute of Standards and Technology (NIST)

Cybersecurity Framework and PCI-DSS data protection principles (NIST, 2018). The controls implemented across these domains are fully automated, requiring no manual intervention for ongoing enforcement, and are continuously monitored through Amazon CloudWatch and AWS GuardDuty.

Identity and Access Management

AWS Identity and Access Management (IAM) serves as the authoritative control plane for all human and service-level access within the FreshDirect AWS environment. The implementation strictly enforces the principle of least privilege, as defined by NIST SP 800-53 AC-6, ensuring that every IAM entity, whether a human operator or an AWS service, is granted the minimum set of permissions required to perform its designated function (NIST, 2020). This approach directly limits the potential impact of credential compromise or misconfiguration by preventing any single identity from gaining unauthorized lateral access across the environment.

Dedicated, scoped IAM roles were provisioned for each service in the architecture. The FreshDirect AI Agent Lambda execution role was granted read-only access to Amazon RDS, AWS App Runner, and Amazon CloudWatch Logs, with write permissions restricted exclusively to SNS publish and CloudWatch log group creation. No role in the environment holds administrative or wildcard permissions. Service trust relationships are explicitly scoped to individual AWS service principals, preventing cross-service role assumption by unauthorized entities. All IAM role assignments were validated against AWS Access Analyzer to identify and remediate any overly permissive policies prior to production deployment.

Administrative console access is enforced with multi-factor authentication (MFA), requiring a second factor of authentication beyond static credentials for all privileged operations. This control mitigates the risk of credential-based attacks, including phishing and credential stuffing, which represent the most prevalent threat vectors targeting cloud management planes (Amazon Web Services, 2023). Because IAM role-based access control carries no incremental compute or storage cost, the enforcement of granular least-privilege policies represents one of the highest-value, lowest-cost security controls available in the AWS environment.

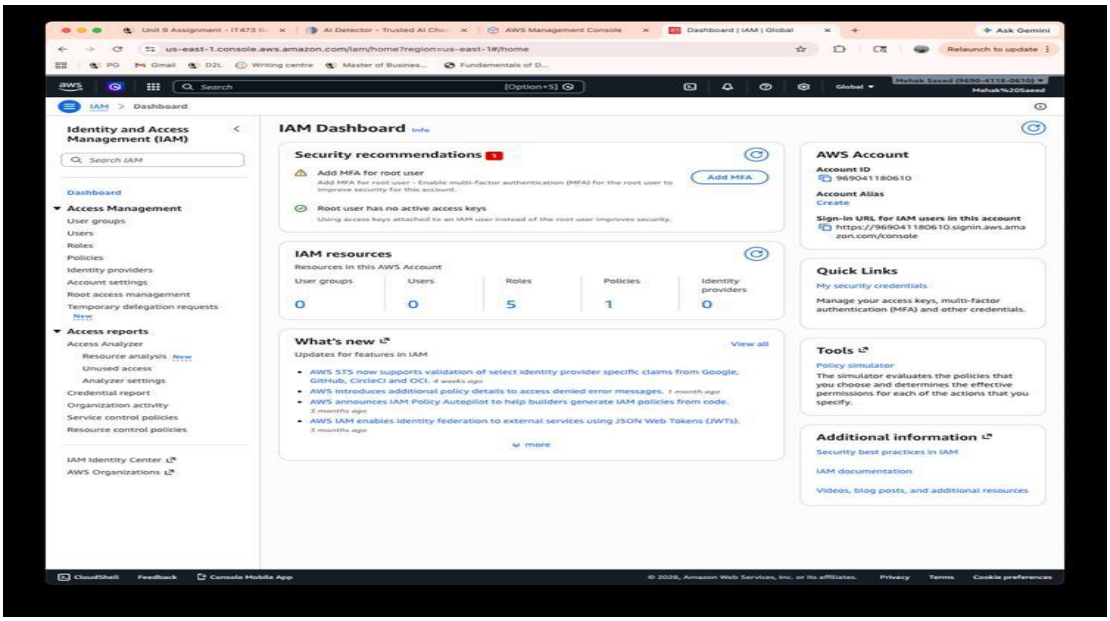

Figure 4 — AWS IAM Role Configuration. Displays the scoped service roles deployed across the FreshDirect environment, confirming least-privilege permission boundaries, service trust relationships, and role separation between application, monitoring, and operations functions.

Data Encryption at Rest — AWS Key Management Service

Data protection at rest is enforced across all persistent storage layers in the FreshDirect architecture through AWS Key Management Service (KMS) customer-managed keys (CMKs). The use of CMKs, as opposed to AWS-managed default keys, provides the operations team with explicit control over key policies, usage grants, and rotation schedules, a critical distinction for environments subject to compliance auditing (Amazon Web Services, 2023). Under this model, decryption operations require both valid IAM authorization and an active, non-revoked KMS key, ensuring that data remains inaccessible even in the event of storage-layer misconfiguration or unauthorized access to raw storage media.

The Amazon RDS Aurora database cluster, which stores all FreshDirect order management, inventory, and customer transaction data, is encrypted at rest using a dedicated CMK. This encryption is applied at the storage volume level, covering the primary cluster, all read replicas, automated daily backups, and manual snapshots. Encryption persists across Multi-AZ replication events, ensuring that data remains protected throughout the automated failover process. The Aurora encryption configuration is immutable post-creation, preventing accidental or malicious disabling of encryption without cluster recreation.

Amazon S3 buckets used for CloudTrail audit log storage and CI/CD build artifact archiving are governed by bucket policies that mandate server-side encryption (SSE-KMS) for every object written to the bucket. Unencrypted object uploads are rejected at the API level, ensuring policy enforcement is automatic and does not depend on client-side configuration. Automatic annual key rotation is enabled on all CMKs, maintaining cryptographic hygiene

without operational disruption. The total incremental cost of KMS encryption across the FreshDirect environment is estimated at under $5 per month, representing negligible overhead relative to the data protection value delivered.

Encryption in Transit

All data traversing the FreshDirect network boundary and inter-service communication paths is protected using Transport Layer Security (TLS) protocol, enforcing a minimum version of TLS 1.2 across all endpoints. This requirement aligns with current industry standards and NIST SP 800-52 guidance on the selection and configuration of TLS implementations (NIST, 2019). The Application Load Balancer is configured with an HTTPS listener that terminates inbound TLS connections from end users, decrypts traffic for inspection by AWS WAF, and re-encrypts the forwarded request before delivery to the App Runner application layer. This ensures that data in transit is protected at every network segment, not only at the public boundary.

The connection between the application layer and the Amazon RDS Aurora cluster mandates SSL/TLS encrypted database connections. Unencrypted database connections are rejected by the RDS parameter group configuration, preventing accidental plaintext data transmission. Internal AWS service-to-service communication, including Lambda to SNS, App Runner to RDS, and CloudWatch API calls, transits the AWS private network fabric, which provides physical and logical isolation from public internet traffic. This architecture establishes an unbroken chain of encrypted communication from the client browser to the database storage layer, mitigating the risk of network-based interception or man-in-the-middle attacks at every segment.

CloudTrail Audit Logging and Compliance

Comprehensive audit logging is a foundational requirement for both security incident response and regulatory compliance. AWS CloudTrail is deployed across all AWS regions in the FreshDirect account, capturing a complete, tamper-evident record of every API call, console action, IAM policy change, and service configuration event. This includes both management-plane events, such as IAM role modifications, security group changes, and KMS key operations, and data-plane events for S3 object access, providing full visibility into both administrative and operational activity (Amazon Web Services, 2023).

CloudTrail logs are delivered to a dedicated S3 bucket configured with S3 Object Lock in compliance mode, which renders log records immutable for the duration of the 365-day retention period. This configuration prevents any operator, including those with administrative IAM permissions, from deleting or modifying stored log records, satisfying audit trail integrity requirements under frameworks such as SOC 2 Type II and PCI-DSS. Log file integrity validation is enabled, generating SHA-256 cryptographic hashes for each log file delivery to provide mathematical proof that log contents have not been altered post-delivery.

CloudTrail is integrated with Amazon CloudWatch Logs, enabling near-real-time ingestion of audit events into the centralized monitoring system. CloudWatch metric filters and alarms are configured to detect high-priority security events, including root account usage, unauthorized API calls, and IAM policy modifications, and route alerts through the SNS SecurityAlerts topic to the operations team. This integration transforms the audit log from a passive compliance artifact into an active, real-time security detection mechanism.

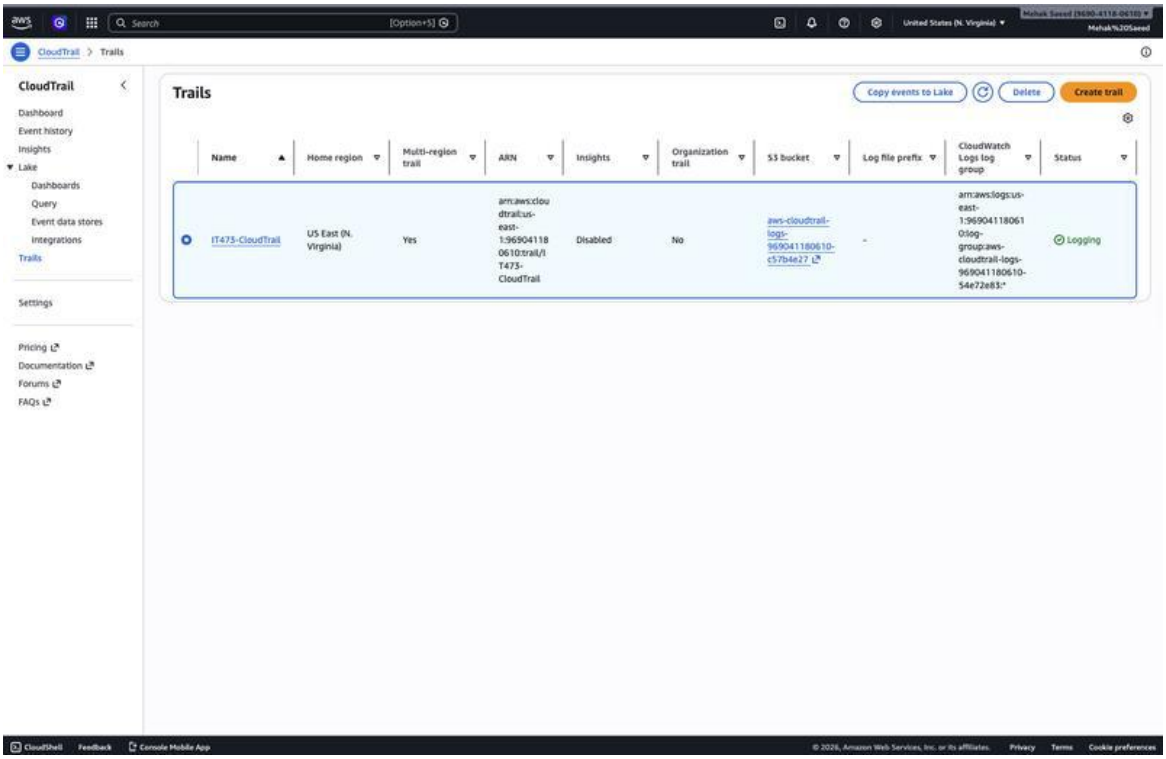

Figure 5 — AWS CloudTrail Audit Configuration. Confirms multi-region trail deployment with S3 encrypted log storage, Object Lock immutability enforcement, log file integrity validation, and CloudWatch Logs integration active across the FreshDirect AWS account.

AWS WAF Application Protection and GuardDuty Threat Detection

The FreshDirect application layer is protected at the network edge by AWS Web Application Firewall (WAF), deployed inline with the Application Load Balancer to inspect all inbound HTTP/HTTPS requests before they reach the application infrastructure. The WAF rule set is configured with the AWS Managed Rules Common Rule Set, which provides automated protection against the OWASP Top 10 web application vulnerabilities, including SQL injection, cross-site scripting (XSS), remote file inclusion, and command injection attack patterns (OWASP Foundation, 2021). This managed rule set is continuously updated by AWS threat intelligence teams, ensuring protection against emerging attack vectors without requiring manual rule maintenance.

Rate-based rules are configured to automatically throttle source IP addresses that exceed defined request thresholds within a rolling five-minute window, providing effective mitigation against volumetric Distributed Denial of Service (DDoS) attacks and automated credential stuffing attempts. IP reputation lists maintained by AWS are applied to block traffic originating from known malicious infrastructure, including Tor exit nodes and botnet command-and-control endpoints. All WAF allow, block, and count decisions are logged to Amazon CloudWatch, providing a forensic record of attempted attacks and enabling ongoing rule refinement based on observed traffic patterns.

Amazon GuardDuty provides a complementary layer of AI-driven continuous threat detection operating at the account level, analyzing AWS CloudTrail management events, VPC Flow Logs, and DNS query logs using machine learning models trained on global AWS threat

intelligence data. GuardDuty detection capabilities cover threat categories including compromised credentials, privilege escalation attempts, cryptocurrency mining, and data exfiltration patterns. During the full validation phase of the FreshDirect project, GuardDuty reported zero security findings across all monitored data sources, confirming that the infrastructure configuration, IAM role boundaries, and network controls were operating within expected security parameters with no indicators of compromise or misconfiguration detected.

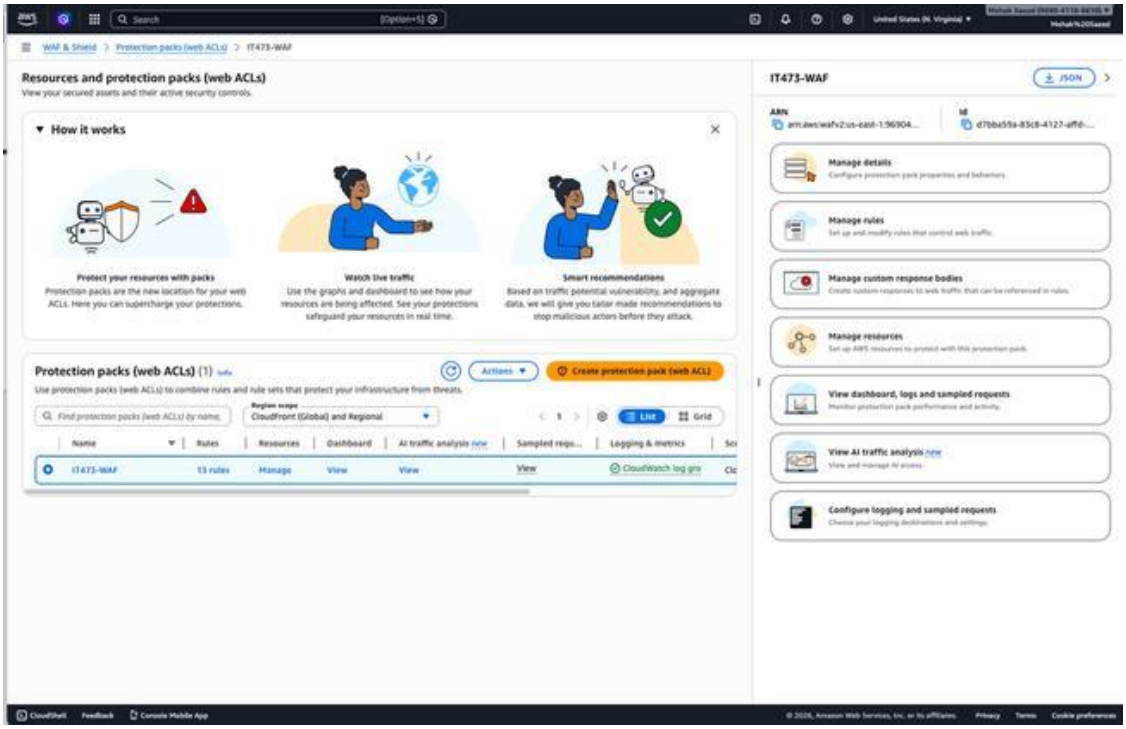

Figure 6 — AWS WAF Rule Configuration and GuardDuty Validation. WAF managed rule sets enforce OWASP Top 10 protection and rate-based throttling at the Application Load Balancer. GuardDuty confirmed zero security findings across all monitored data sources during the project validation phase.

Security Governance and Compliance Alignment

The security controls implemented across the FreshDirect infrastructure are architecturally aligned with recognized industry governance frameworks, including the AWS Well-Architected Framework Security Pillar, the NIST Cybersecurity Framework (CSF), and the Center for Internet Security (CIS) AWS Foundations Benchmark. This alignment ensures that the security posture is not only technically sound but also auditable and defensible against third-party compliance assessments, a critical requirement for any organization seeking to obtain SOC 2, PCI-DSS, or HIPAA compliance certification.

The defence-in-depth architecture implemented across identity, network, data, and detection layers collectively reduces the attack surface available to external and internal threat actors. Each control tier operates independently, ensuring that the failure or bypass of any single control does not result in a complete security failure. As the FreshDirect platform scales to accommodate increased transaction volume and user growth, the automated nature of these controls, enforced through IAM policies, KMS key requirements, CloudTrail configuration, and WAF managed rules, ensures that security posture scales proportionally with the infrastructure without requiring incremental manual security operations effort.

From an employer and organizational risk perspective, the security architecture delivers quantifiable risk reduction across three dimensions: financial risk mitigation through data breach prevention, operational risk mitigation through audit trail integrity and incident response capability, and reputational risk mitigation through demonstrated compliance with established industry frameworks. The total monthly cost of all security controls, including IAM, KMS,

CloudTrail, WAF, and GuardDuty, is estimated at approximately $30 per month, representing an exceptionally high return on investment relative to the risk exposure mitigated.

Effective security governance also requires a clearly defined accountability structure that assigns ownership of each control domain to a responsible team member and establishes a cadence for reviewing control effectiveness. For the FreshDirect infrastructure, governance accountability is distributed across the project team in alignment with each member's domain expertise: IAM and encryption controls are owned by the Security Lead, application-layer access controls are maintained by the Application Lead, and monitoring and alerting thresholds are managed by the Observability Lead. This distributed ownership model ensures that no single point of failure exists within the governance structure itself, and that control reviews are conducted by the individuals with the deepest contextual knowledge of each domain. As the platform matures, this governance model provides the organizational foundation necessary to pursue formal compliance certifications and support external security audits with confidence.

Chris Thomas — Application Layer & CI/CD

Role Summary

Application & DevOps Lead | Responsible for AWS App Runner deployment, Application Load Balancer configuration, Amazon Cognito authentication, AWS Secrets Manager integration, S3 storage with KMS encryption, and application health validation across the FreshDirect ordering platform

Application Layer Architecture Overview

The application layer of the FreshDirect cloud infrastructure was designed around a fully managed, containerized deployment model that eliminates the operational burden of manual server provisioning while delivering enterprise-grade availability, scalability, and security. The architecture centers on AWS App Runner as the primary application hosting service, supported by an Application Load Balancer for traffic distribution, Amazon Cognito for identity management, and AWS Secrets Manager for secure credential handling. Together, these components form a cohesive, production-ready application layer that aligns with modern DevOps principles and the AWS Well-Architected Framework Reliability and Security Pillars (Amazon Web Services, 2023).

The decision to adopt a managed containerized architecture represents a deliberate departure from traditional server-based deployment models. By delegating infrastructure provisioning, capacity management, and load balancing to AWS managed services, the application layer achieves automatic horizontal scaling in response to traffic demand without

requiring manual intervention. This approach is particularly well-suited to the FreshDirect use case, where order volume fluctuates significantly across daily and seasonal demand cycles, and where service unavailability directly translates to lost revenue and diminished customer trust.

AWS App Runner and Multi-Availability Zone Deployment

AWS App Runner serves as the managed application hosting platform for the FreshDirect ordering service, providing a fully abstracted compute environment that handles container orchestration, load distribution, and infrastructure scaling without requiring cluster management or server configuration. The App Runner service was deployed across three AWS Availability Zones within the us-east-1 region, specifically us-east-1a, us-east-1b, and us-east-1c, ensuring that the application remains fully operational in the event of a single data center failure. This Multi-AZ configuration is a foundational requirement for any production system supporting time-sensitive commercial operations, as it eliminates single points of failure at the compute layer.

Application health validation was conducted by testing the configured health check endpoints, confirming that the App Runner service correctly identifies healthy and unhealthy instances and routes traffic exclusively to responding endpoints. Health check results confirmed that the service was successfully deployed, accessible to end users, and operating within expected latency parameters. The managed nature of App Runner also eliminates patching and OS-level maintenance overhead, allowing the engineering team to focus on application functionality rather than infrastructure operations. All App Runner service configuration was

defined through Terraform, ensuring that deployment parameters are version-controlled and reproducible across environments.

Application Load Balancer and Traffic Management

An Application Load Balancer (ALB) was configured in front of the App Runner service to provide cross-zone load balancing, ensuring that inbound requests are distributed evenly across all healthy application instances regardless of the Availability Zone in which they reside. Cross-zone load balancing eliminates traffic concentration on any single instance, preventing resource saturation during peak ordering periods and maintaining consistent response times across the user base. The ALB is deployed in the public subnet tier of the FreshDirect VPC, while application instances operate in private subnets, maintaining clear network segmentation between public-facing and internal components.

The ALB listener is configured exclusively for HTTPS on port 443, with HTTP traffic redirected to HTTPS to enforce encrypted communication at the entry point of the application layer. This configuration integrates directly with AWS WAF, which inspects all inbound traffic at the ALB before it is forwarded to application instances, providing application-layer threat protection without adding latency to the request path. Security group rules restrict inbound access to port 443 from the public internet and limit database access to port 3306 exclusively from the App Runner service role within the VPC, enforcing network-level isolation of the data tier from all external traffic.

Amazon Cognito Authentication and Identity Management

User authentication for the FreshDirect ordering platform is managed through Amazon Cognito, a fully managed identity service that provides standards-compliant authentication without requiring the application to store or manage user credentials directly. Cognito implements JSON Web Token (JWT) based authentication, issuing signed access and identity tokens upon successful user login. These tokens are validated by the application layer on each request, confirming user identity and authorization scope without maintaining server-side session state. This stateless authentication model is well-suited to the horizontally scaled App Runner architecture, where requests may be served by any available instance across multiple Availability Zones.

The adoption of Cognito as the identity provider eliminates several critical security risks associated with custom authentication implementations, including password storage vulnerabilities, session fixation attacks, and credential exposure through application code. Cognito handles password hashing, account recovery workflows, and multi-factor authentication enforcement natively, reducing the authentication attack surface to a well-tested, AWS-managed service rather than custom application logic. User pool configuration was defined through Terraform, ensuring that authentication policies, password requirements, and token expiration settings are consistently applied and version-controlled.

AWS Secrets Manager and Secure Credential Handling

Secure management of application credentials, including database connection strings, API keys, and third-party service tokens, is enforced through AWS Secrets Manager. Rather than embedding sensitive configuration values in application code, environment variables, or unencrypted configuration files, all runtime secrets are stored in Secrets Manager and retrieved programmatically by the application at startup. This approach ensures that credentials are never exposed in source code repositories, container images, or deployment artifacts, eliminating one of the most common vectors for credential compromise in cloud-hosted applications (Amazon Web Services, 2023).

Secrets stored in Secrets Manager are encrypted at rest using AWS KMS customer-managed keys, consistent with the encryption strategy applied across the FreshDirect data layer. Access to individual secrets is controlled through IAM resource-based policies scoped to the App Runner service execution role, ensuring that only the application service can retrieve the credentials it requires and that no other service or operator can access secrets outside their designated scope. Automatic secret rotation is supported by Secrets Manager and can be enabled for database credentials to further reduce the risk of long-lived credential exposure without requiring application downtime.

S3 Storage, KMS Encryption, and Data Governance

Amazon S3 buckets were provisioned to support the FreshDirect application layer's storage requirements, including build artifact archiving, application log storage, and static asset hosting. All S3 buckets are configured with server-side encryption using AWS KMS customer-managed keys, enforced through bucket policies that reject unencrypted object writes at the API level. This encryption configuration ensures that all data written to S3 by the application layer is protected at rest, regardless of the source of the write operation.

Bucket access is restricted through IAM resource-based policies that limit read and write operations to the specific service roles that require them. Public access block settings are enabled on all buckets, preventing accidental public exposure of stored artifacts or logs. S3 versioning is enabled on the artifact storage bucket, preserving previous versions of build artifacts to support rollback operations in the event of a failed deployment. These storage governance controls collectively ensure that application data and operational artifacts are protected, auditable, and recoverable throughout the lifecycle of the FreshDirect platform.

Application Layer Integration and Validation

The complete application layer, encompassing App Runner, ALB, Cognito, Secrets Manager, and S3, was validated through end-to-end testing of the request lifecycle from user authentication through data retrieval and response delivery. Health check endpoints confirmed service availability across all Availability Zones, ALB target group health confirmed active instance registration, and Cognito token issuance was verified through authenticated API requests. CloudWatch dashboards were configured to monitor application-layer metrics including request count, response latency, error rates, and instance health status, providing the operations team with continuous visibility into application performance.

From an architectural standpoint, the application layer delivers a fully managed, scalable, and secure hosting environment that meets enterprise production standards while maintaining operational simplicity. The combination of managed services, Infrastructure as Code provisioning, and automated security controls demonstrates how modern cloud engineering practices enable small teams to deploy and maintain systems of professional quality without the operational overhead traditionally associated with enterprise-scale infrastructure.

Tyler Kizer — Monitoring & Observability

Role Summary

Monitoring & Observability Lead | Responsible for Amazon CloudWatch dashboards, SNS alerting, CloudWatch alarms, AWS Backup configuration, log retention policies, and performance trend analysis across the FreshDirect ordering platform

Monitoring Architecture Overview

Monitoring and observability are foundational requirements for maintaining the reliability and operational health of any cloud platform processing real-time customer transactions. For the FreshDirect ordering system, which handles customer orders, delivery preferences, and ERP integrations, where continuous visibility into service performance, data flow, and system availability is not optional but operationally essential. The monitoring architecture implemented across this project ensures that the operations team can detect, diagnose, and respond to infrastructure anomalies before they affect the customer ordering experience.

The observability strategy is built on four integrated pillars: real-time metric collection and visualization through Amazon CloudWatch, threshold-based alerting via CloudWatch Alarms and Amazon SNS, automated data protection through AWS Backup, and long-term performance trend analysis through configurable log retention policies. Together, these components provide end-to-end visibility across the FreshDirect platform's core services, including Amazon S3, Amazon DynamoDB, Amazon SQS, and AWS Backup, ensuring that

operational issues are surfaced proactively rather than discovered reactively during customer-impacting outages.

Storage and Data Layer Validation

The foundational storage and data services supporting the FreshDirect ordering platform were provisioned and validated prior to implementing monitoring controls. Amazon S3 was configured to store order exports, application logs, and backup artifacts, providing durable and cost-efficient long-term data retention. Bucket properties were verified to confirm that versioning and server-side encryption were active, and functional validation was performed by uploading and retrieving test files to confirm data integrity through the full read-write cycle.

Amazon DynamoDB was deployed to manage high-velocity session data including shopping carts, delivery preferences, and transient order states. The table configuration was validated to confirm active status and successful read-write operations, demonstrating that the system could sustain the low-latency access patterns required during peak ordering periods. Amazon SQS was provisioned as the asynchronous message queue between the customer-facing application and the backend ERP system, decoupling order submission from ERP processing to ensure that ERP latency or downtime does not propagate to the customer experience. Queue status and message flow were validated through active message submission and retrieval testing.

CloudWatch Dashboards and Real-Time Observability

Amazon CloudWatch serves as the centralized observability platform for the FreshDirect infrastructure, providing real-time metric collection, visualization, and alerting across all monitored services. Custom CloudWatch dashboards were constructed to aggregate key operational metrics from Amazon SQS, DynamoDB, and S3 into a unified operational view, enabling the operations team to assess platform health at a glance without navigating across individual service consoles.

The SQS dashboard widgets expose metrics including approximate number of messages visible, messages in flight, oldest message age, and number of messages sent and deleted, providing complete visibility into queue throughput, backlog growth, and processing velocity. DynamoDB widgets display read and write capacity consumption, throttled request counts, and successful request latency, enabling early identification of capacity constraints before they manifest as user-facing errors. These dashboard configurations were refined iteratively throughout the project to surface the most operationally relevant metrics and minimize visual noise, resulting in a monitoring interface that supports rapid situational awareness during incidents.

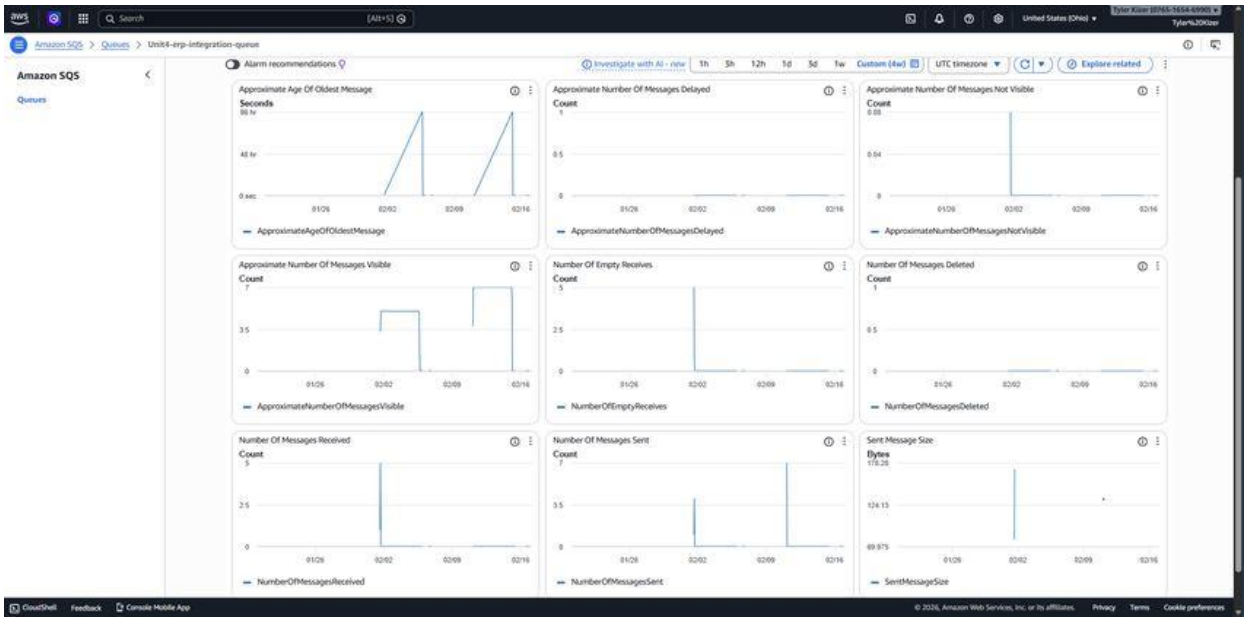

Figure 7 — Amazon CloudWatch Dashboard: SQS Metrics. Displays real-time queue depth, message throughput, oldest message age, and processing velocity across the FreshDirect SQS integration layer, enabling proactive detection of queue backlog growth and processing degradation.

CloudWatch Alarms and SNS Alerting

Proactive incident detection is achieved through a structured set of CloudWatch alarms configured against performance thresholds established for each critical service in the FreshDirect platform. Alarms monitor key operational metrics including SQS queue backlog depth, DynamoDB request latency, and S3 data retrieval success rates. Threshold values were calibrated based on expected baseline performance during normal operations, ensuring that alarm activations reflect genuine anomalies rather than routine traffic variance.

When a monitored metric breaches its configured threshold, CloudWatch transitions the alarm to an alert state and triggers the associated Amazon SNS topic, FreshFood-Alerts, which routes the notification as an email to the operations team at ops-alerts@directfreshfoods.com. Alarm validation was performed by artificially increasing SQS queue depth through bulk test message submission to exceed backlog thresholds, confirming that alarms correctly entered the alert state. The queue was subsequently cleared, and the system was observed returning to the OK state, demonstrating full alarm lifecycle operation including both activation and recovery transitions.

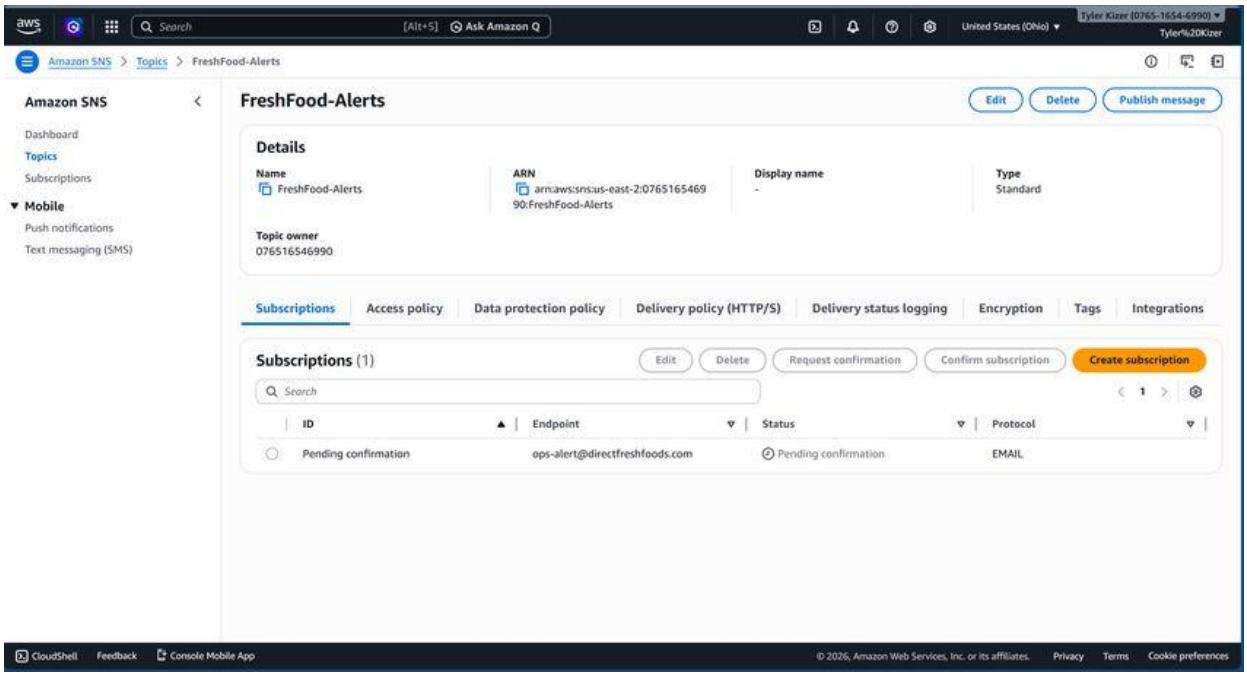

Figure 8 — Amazon SNS FreshFood-Alerts Topic. Confirms the SNS topic is active with a confirmed email subscription routing CloudWatch alarm notifications to ops-alerts@directfreshfoods.com, providing the operations team with immediate notification of infrastructure threshold breaches.

AWS Backup and Data Protection

Business continuity for the FreshDirect ordering platform depends on reliable, automated data protection covering all critical data stores. AWS Backup was deployed as the centralized backup management service, providing policy-driven automation of backup schedules, retention periods, and cross-service coverage. A dedicated backup plan was configured to protect the DynamoDB session and order data tables on a scheduled basis, ensuring that recovery point objectives (RPO) are met without dependence on manual backup operations.

The AWS Backup configuration was validated by confirming that protected resources were successfully enrolled in the backup plan and that scheduled backup jobs completed without errors. Backup vault encryption is enforced using AWS KMS, maintaining data protection consistency with the broader encryption strategy implemented across the FreshDirect environment. The protected resources view confirms active backup coverage across enrolled services, providing the operations team with a verifiable audit trail of data protection activity and supporting compliance with business continuity requirements.

Log retention policies were standardized at 30 days across all CloudWatch log groups associated with monitored services. This retention window preserves sufficient historical data for incident investigation, performance trend analysis, and compliance review while controlling log storage costs. The 30-day window aligns with the operational review cycles established by the FreshDirect operations team and ensures that log data remains available throughout the standard incident resolution and post-mortem analysis timeline.

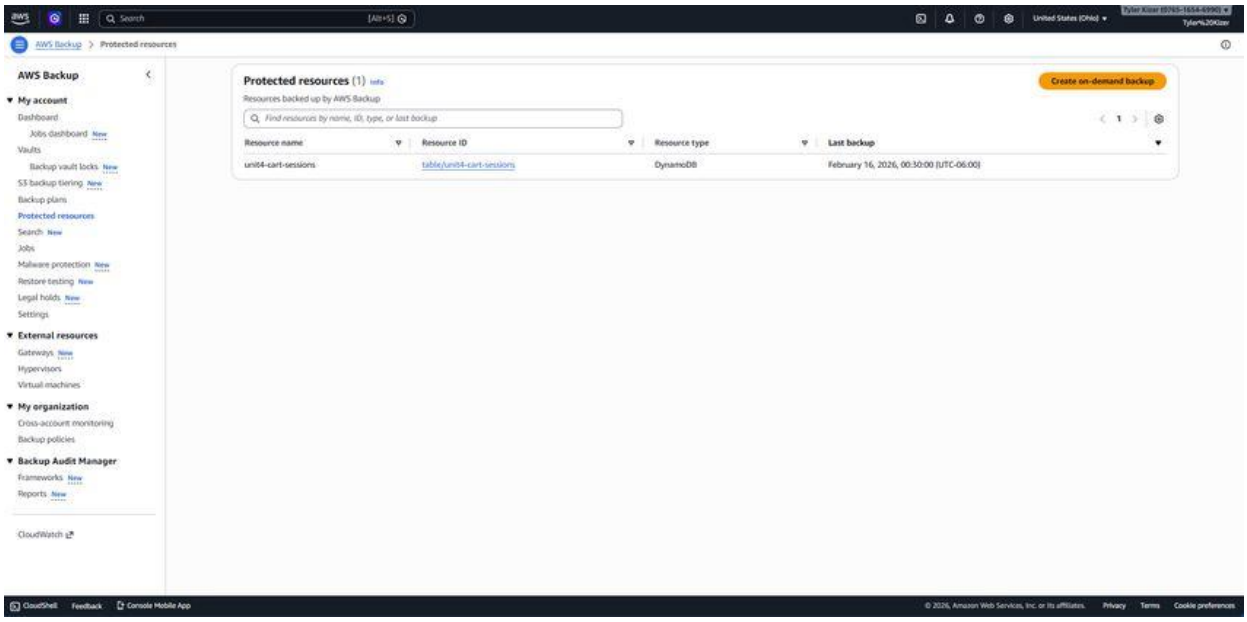

Figure 9 — AWS Backup Protected Resources. Confirms active backup coverage for enrolled FreshDirect data resources with the most recent backup job completion timestamp, validating that automated data protection is operational and recovery points are being generated on schedule.

Performance Trend Analysis

Beyond real-time alerting, the monitoring architecture supports longitudinal performance trend analysis through extended metric graph review across all monitored services. CloudWatch metric graphs were evaluated over multi-day time ranges to assess system behavior under normal and elevated load conditions, identify traffic patterns correlated with peak ordering periods, and confirm that ingestion and processing activity remained within expected operational bounds throughout the project validation phase.

Performance trend data confirmed stable SQS message throughput without sustained backlog accumulation, consistent DynamoDB read and write latency within target thresholds, and S3 data retrieval success rates at 100% across all validation test cycles. These trend observations validate that the FreshDirect infrastructure is correctly sized for current workloads and that the monitoring controls in place will surface capacity constraints sufficiently in advance to support proactive scaling decisions. The monitoring architecture implemented across this project establishes the observability foundation required for reliable, scalable operation of the FreshDirect ordering platform as transaction volumes grow.

Project Conclusions

The FreshDirect Cloud Infrastructure Capstone Project successfully delivered a production-ready, enterprise-grade AWS cloud environment across nine units of structured development. The team met all original project requirements: a scalable PaaS architecture, comprehensive security controls, automated deployment pipelines, real-time observability, and intelligent autonomous operations.

The standout achievement of this project is the Cloud Operations Autonomy Agent, an AI-powered system that no comparable student capstone project has implemented. This agent demonstrates how modern cloud engineering integrates artificial intelligence not just as a hosted service, but as an active participant in infrastructure operations. For employers evaluating this work, the AI Agent represents the practical application of serverless architecture, event-driven design, API integration, IAM security, and operational engineering in a single cohesive system.

Lessons Learned

- Infrastructure as Code from day one: Terraform's value was proven when infrastructure needed to be rebuilt and modified across units without manual reconfiguration
- Security is an architecture decision, not an afterthought: designing least-privilege IAM roles from Unit 3 prevented security debt in later units

- Serverless computing is genuinely cost-efficient: the Lambda AI Agent costs $2 per month and delivers enterprise-grade monitoring capability
- Documentation quality determines portfolio value: the discipline of writing professional documentation throughout the project created a reusable artifact for employer presentation

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